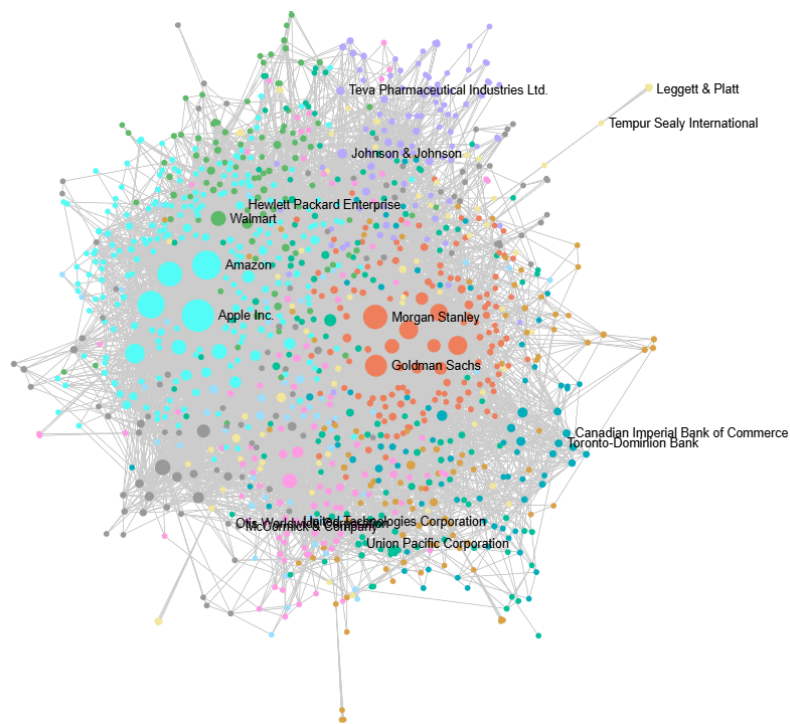


Company Co-Mention Graphs in News Articles

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1 Introduction and Project Context

This project investigates how publicly listed companies are represented together in financial news articles. The main objective is to transform a large collection of unstructured financial news articles into a company co-mention network and then study whether this network reflects meaningful economic structure, including industry groups, company size, and financial behaviour.

In the co-mention network, each node represents a company and each weighted edge represents the number of articles in which two companies are mentioned together. The assumption behind this approach is that companies mentioned in the same article may be connected by a shared market narrative, sector relationship, competitive relationship, supply-chain link, regulatory event, macroeconomic shock, or investor attention pattern.

2 Data Sources and Company Reference Dataset

This section describes the two main data sources used in the project: the financial news article dataset and the Wikidata company reference dataset. It also explains how the raw company reference data was cleaned and standardized before being used for entity matching.

2.1 Financial News Articles

The financial news dataset was generated from the Hugging Face dataset:

`ashraq/financial-news-articles`. The dataset was loaded into a pandas dataframe and filtered to keep only articles with an identifiable publication date. Dates were extracted from the first 100 characters of each article using a regular expression designed to identify month-day-year formats between 2010 and 2019.

After extracting publication dates, the analysis was restricted to articles published in 2018. This provided a focused one-year window and avoided mixing different market periods. The dataset was then cleaned by removing clearly non-financial articles based on URL keywords. The exclusion list included topics such as lifestyle, travel, weather and geopolitical content.

After these filters, a random sample of 50,000 financial articles were selected.

2.2 Wikidata Company Reference Dataset

A structured company reference dataset was created from Wikidata using a SPARQL query. This dataset was needed because the raw organization names extracted from articles had to be matched to standardized company records. The query selected companies listed on either the New York Stock Exchange or Nasdaq.

For each company, the query retrieved the company entity, company label, Wikidata company ID, exchange label, ticker symbol, industry labels, market capitalization, market capitalization date, revenue, and revenue date. Multiple industry labels were combined into one semicolon-separated field using `GROUP_CONCAT`.

Listing 1: SPARQL query used to retrieve NYSE and Nasdaq companies from Wikidata.

```

1 SELECT ?company ?companyLabel ?companyId ?exchangeLabel
2     (SAMPLE(?ticker0) AS ?ticker)
3     (GROUP_CONCAT(DISTINCT ?industryLabel0; separator="; ") AS ?industries)
4     (SAMPLE(?marketCap0) AS ?marketCap)
5     (SAMPLE(?marketCapDate0) AS ?marketCapDate)
6     (SAMPLE(?revenue0) AS ?revenue)
7     (SAMPLE(?revenueDate0) AS ?revenueDate)
8 WHERE {
9     ...
10 }
11 GROUP BY ?company ?companyLabel ?companyId ?exchangeLabel
12 ORDER BY ?companyLabel
13 LIMIT 4500

```

2.3 Cleaning and Standardization

The exported SPARQL result was cleaned into a usable company reference table. The raw Wikidata output contained 4,268 rows. After grouping duplicate company identifiers, the cleaned reference dataset contained 4,189 unique companies.

The cleaning process standardized company IDs as strings, stripped whitespace from company names and tickers, converted invalid values such as empty strings, **nan**, and **None** into missing values, converted revenue and market capitalization into numeric fields, and converted financial dates into datetime format.

Because some companies appeared multiple times due to multiple exchange listings, the data was grouped by `company_id`. For each company, the first available company name, revenue, market capitalization, and URL were retained, while industries were combined.

The cleaned company reference dataset contained 606 companies with revenue data, 215 companies with Wikidata market capitalization data, 4,190 unique companies. A final manual correction removed Nasdaq' Inc. corporation, which was unwanted.

Measure	Value
Raw Wikidata rows	4,268
Unique cleaned companies	4,189
Companies with revenue data	606
Companies with Wikidata market capitalization data	215

Table 1: Summary of the cleaned Wikidata company reference dataset.

3 Entity Extraction, Alias Matching, and Co-Mention Edge Construction

This section explains how the project transformed raw article text into matched companies and then into company-company co-mention edges. These steps form the main bridge between unstructured financial news and the final network dataset.

3.1 Organization Extraction with spaCy

Organization extraction was performed using the `en_core_web_sm` spaCy model. Each article was processed with spaCy’s named entity recognition pipeline, and only entities labelled as `ORG` were retained. The `ORG` label captures organizations, companies, institutions, agencies, and other named organizational entities.

This produced a list of organization mentions for each article. However, raw spaCy output can include many entities that are not target companies, such as news agencies, publication names, reporting labels, and generic business phrases. Therefore, additional cleaning was required before company matching.

3.2 Entity Cleaning and Alias Normalization

Before matching, extracted organization names and Wikidata company names were converted into normalized aliases. The normalization removed common formatting differences such as capitalization, possessive endings, punctuation, ampersands, ticker suffixes, repeated spaces, and legal suffixes such as Inc., Corp., Ltd...

Obvious non-company entities, including news agencies, media outlets, and generic organization phrases, were excluded. The resulting normalized aliases were then used for exact matching between article entities and the Wikidata company reference dataset.

3.3 Company Matching

An alias table was constructed from the cleaned Wikidata company reference dataset. Aliases were generated from company names and ticker symbols. The initial alias table contained 7,792 aliases. However, some aliases were ambiguous because they pointed to more than one company, removing them left us with 7,343 clean aliases.

Each article’s cleaned organization entities were normalized and compared with the clean alias dictionary. If a normalized organization name matched an alias, the corresponding Wikidata company ID was assigned to the article. Duplicate company IDs within the same article were removed so that a company was counted only once per article.

The matching process produced 17,597 articles with at least one matched company and 6,081 articles with at least two matched companies. Since a co-mention edge requires at least two companies

in the same article, only the latter group could be used for edge construction. Across all matched articles, 2,562 unique companies were identified.

3.4 Edge Construction

For every article containing at least two matched companies, all unordered company pairs were generated. Each pair received an edge weight equal to the number of articles in which the two companies appeared together. Formally, the edge weight between companies i and j is:

$$w_{ij} = \sum_{a=1}^N \mathbf{1}\{i \in a \text{ and } j \in a\}, \quad (1)$$

where N is the number of articles.

Although 2,562 unique companies were matched at the article level, the initial graph contained 1,898 nodes. This difference occurs because not every matched company necessarily appears in a valid co-mention edge after the pair construction step. Companies mentioned only in single-company articles, or companies not connected to another matched firm in the edge list, do not become nodes in the co-mention graph.

The initial graph contained 1,898 nodes and 9,092 weighted edges. The strongest co-mention pairs included Apple–Google, Apple–Amazon, Microsoft–Apple, Goldman Sachs–Morgan Stanley, Microsoft–Amazon, Citigroup–Morgan Stanley, and Amazon–Walmart.

4 Co-Mention Graph Analysis and Centrality

This section analyses the structure of the co-mention graph using graph filtering and centrality measures. The goal is to identify which companies occupy the most important positions in the financial news network.

4.1 Graph Construction and Filtering

The co-mention edge table was loaded into NetworkX as an undirected weighted graph. In this graph, companies are nodes and co-mention counts are edge weights. The initial graph contained 1,898 nodes and 9,092 edges.

To reduce noise from companies that appeared only weakly in the network, nodes with degree lower than 3 were removed. After this filtering step, the graph contained 1,172 nodes and 8,166 edges. This filtered graph was used for most of the centrality analysis and community detection.

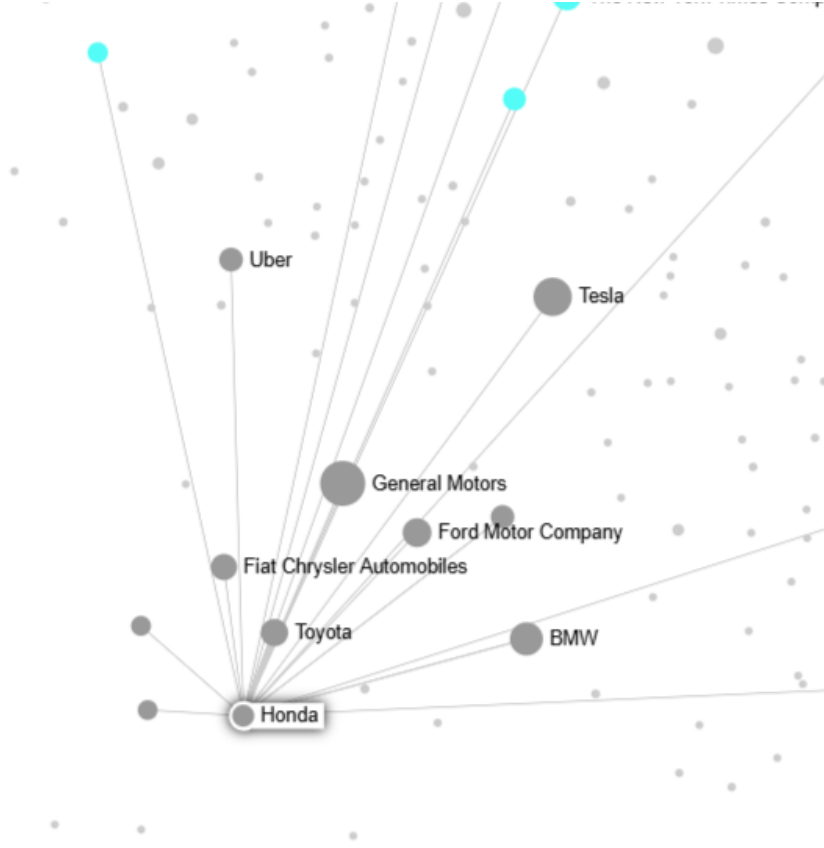


Figure 1: Local neighbourhood of Honda in the co-mention graph.

4.2 Weighted Degree, PageRank, and Betweenness

Several centrality measures were computed. Degree measures the number of distinct companies connected to a company. Weighted degree sums the edge weights connected to a company and therefore captures both the number and frequency of co-mentions. For a company i , weighted degree is:

$$k_i^w = \sum_j w_{ij}, \quad (2)$$

where w_{ij} is the co-mention weight between companies i and j .

PageRank was calculated using edge weights. In this context, PageRank identifies companies that are connected to other important companies. A company can therefore receive a high PageRank score not only by having many co-mentions, but also by being connected to other central companies. Betweenness centrality was used to identify companies that act as bridges between different parts of the network. Since stronger co-mentions represent closer relationships, edge distance was defined as the inverse of edge weight:

$$d_{ij} = \frac{1}{w_{ij}}. \quad (3)$$

This means that companies that are frequently co-mentioned are treated as closer together when computing shortest paths.

4.3 Central Companies

The most central companies by weighted degree included Apple, Amazon, Google, Microsoft, Morgan Stanley, Goldman Sachs, Alphabet, Citigroup, Deutsche Bank, and Credit Suisse. These firms were central because they appeared frequently with many different companies across the article dataset.

Apple had the highest weighted degree, with a value of 1,126. Amazon followed with 979, Google with 910, Microsoft with 800, and Morgan Stanley with 786. These companies also ranked highly by PageRank and betweenness centrality.

The relationship between weighted degree and betweenness was strong. The Pearson correlation between the two metrics was approximately 0.872, with a p-value below 0.001. Among the top 20 companies by weighted degree and the top 20 companies by betweenness, 14 companies overlapped, giving an overlap share of 70%. This suggests that the companies most frequently co-mentioned in financial news also tend to act as bridges across different parts of the network.

Company	Weighted Degree	Betweenness	PageRank
Apple Inc.	1,126	0.433	0.0234
Amazon	979	0.393	0.0202
Google	910	0.213	0.0182
Microsoft	800	0.187	0.0169
Morgan Stanley	786	0.301	0.0167

Table 2: Examples of highly central companies in the filtered graph.

4.4 Centrality by Industry and Company Size

Centrality was also interpreted by industry and company size. Large technology companies and major financial institutions dominated many of the highest centrality rankings. This reflects the structure of financial news, where companies such as Apple, Amazon, Google, Microsoft, Goldman Sachs, and Morgan Stanley are frequently discussed in relation to broader market movements, earnings announcements, regulation, investment banking activity, and sector-level trends.

Industry differences are important because financial news does not mention all sectors equally. Technology and finance companies are more likely to appear in broad market narratives, while firms in smaller or more specialized sectors may be mentioned in narrower contexts. Similarly, large companies are more likely to receive news coverage and to be connected to many other firms.

5 Louvain Communities, Industry Structure, and Robustness

This section evaluates whether the co-mention graph reflects real-world industry structure. Louvain community detection was used to identify graph communities, and these communities were then

compared with broad industry groups.

5.1 Louvain Community Detection

Louvain community detection was applied to the filtered weighted graph. The algorithm detected 17 communities, with a modularity score of approximately 0.47.

Modularity measures the extent to which nodes are more densely connected within communities than between communities. A modularity value of 0.47 indicates a moderate community structure. This means that the co-mention graph does contain groups of companies that are more strongly connected to each other, although the separation between groups is not perfect.

5.2 Industry Interpretation

To interpret the Louvain communities, detailed Wikidata industry labels were mapped into broader industry groups. These groups included Finance, Energy, Technology/Telecom, Healthcare, etc...

Some companies had multiple possible industry labels. For these cases, we used cluster context to assign a more suitable broad industry group. If a company had several possible categories, the dominant industry groups of its Louvain cluster were used to select the most contextually appropriate classification. This approach was useful for diversified companies such as Amazon that is both in the technology industry and the retail industry.

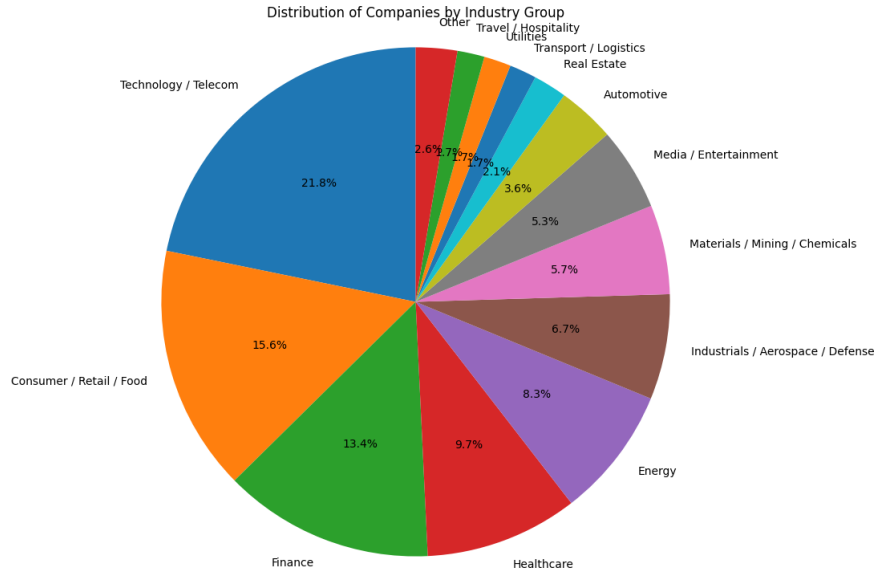


Figure 2: Industry group distribution in the co-mention graph.

5.3 Cluster Purity

Cluster purity was used to evaluate how well Louvain communities aligned with broad industry categories. Companies with unknown or unmapped industry labels were excluded from the purity

calculation. A contingency table was created between Louvain clusters and industry groups. Purity was calculated by taking the dominant industry count in each cluster, summing those dominant counts across all clusters, and dividing by the total number of evaluated companies:

$$\text{Purity} = \frac{1}{N} \sum_{c=1}^C \max_g n_{cg}, \quad (4)$$

where N is the number of evaluated companies, C is the number of clusters, and n_{cg} is the number of companies in cluster c belonging to industry group g .

The overall cluster purity was approximately 0.494. This means that about 49.4% of evaluated companies belonged to the dominant industry group of their detected Louvain cluster. The result suggests that the co-mention graph partially reflects industry structure, but not perfectly.

This partial alignment is expected. Financial news does not group companies only by industry. Articles can also connect companies through macroeconomic themes, regulation, earnings announcements, investor sentiment, mergers and acquisitions, litigation, supply-chain events, or general market news.

5.4 Robustness with Stronger Edge Thresholds

A robustness test was performed by increasing the minimum edge weight required for an edge to remain in the graph. This removed weak one-off co-mentions and kept only stronger repeated relationships. As the edge threshold increased, cluster purity also increased.

With a minimum edge weight of 5, purity increased to approximately 0.73. With thresholds of 10, 15, and 20, purity increased further to approximately 0.869, 0.896, and 0.888 respectively. This shows that repeated co-mentions are more likely to represent meaningful business or industry relationships than single co-mentions.

Minimum Edge Weight	Nodes	Edges	Clusters	Purity
5	116	428	13	0.730
10	46	121	7	0.869
15	29	72	5	0.896
20	18	40	3	0.888

Table 3: Cluster purity under stronger co-mention thresholds.

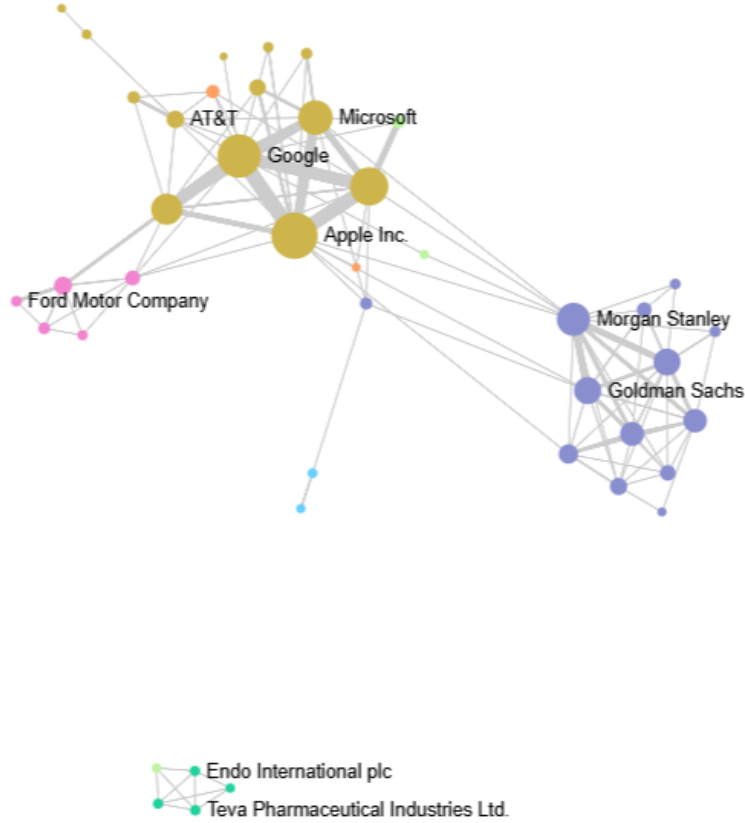


Figure 3: Louvain co-mention graph with minimum edge weight of 10.

Overall, Louvain clustering, industry interpretation, purity, and robustness testing all address the same research question: whether financial-news co-mention communities correspond to real-world industry structure. The results show that there is meaningful alignment, especially when weaker co-mentions are removed.

6 Article Counts, Centrality, and Company Size

The main hypothesis is that larger companies receive more financial news attention and occupy more central positions in the co-mention graph. Company size is measured using revenue and market capitalization. News attention is measured using article counts. Network position is measured mainly using weighted degree.

6.1 Article Counts and Company Size

Ordinary least squares regressions were used to test whether companies with more article mentions tend to be larger. The first model used log revenue as the dependent variable and log article count as the independent variable:

$$\log(\text{Revenue}) = \beta_0 + \beta_1 \log(\text{Article Count}) + \epsilon. \quad (5)$$

This model produced an R^2 of 0.138. The coefficient on log article count was 0.747 and statistically significant, with a p-value of 7.10×10^{-18} and a sample size of 504 companies.

The second model used log market capitalization as the dependent variable:

$$\log(\text{Market Cap}) = \beta_0 + \beta_1 \log(\text{Article Count}) + \epsilon. \quad (6)$$

This model produced an R^2 of 0.352. The coefficient on log article count was 0.914 and statistically significant, with a p-value of 2.73×10^{-16} and a sample size of 157 companies. This suggests that article counts are positively associated with both revenue and market capitalization, with a stronger fit for market capitalization.

Dependent Variable	R^2	Coefficient	p-value	N
Log revenue	0.138	0.747	7.10×10^{-18}	504
Log market capitalization	0.352	0.914	2.73×10^{-16}	157

Table 4: OLS regressions of company size on article counts.

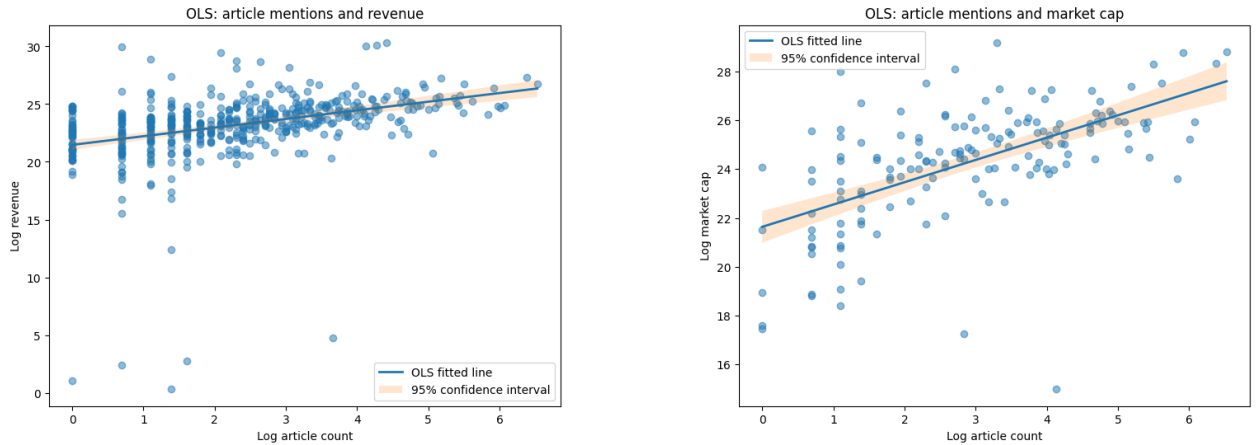


Figure 4: OLS regressions of log revenue and log market capitalization on log article counts.

6.2 Centrality and Company Size

The relationship between company size and weighted degree was also tested. Because both company size and network centrality are skewed, the analysis used log transformations and non-parametric

comparisons across company-size quartiles.

The Pearson correlation between log revenue and log weighted degree was approximately 0.357, with a p-value of 1.998×10^{-12} and a sample size of 365 companies. This indicates a statistically significant positive relationship between revenue and centrality.

Companies were also divided into revenue quartiles. Median weighted degree increased from 7.0 in the lowest revenue quartile to 60.0 in the highest revenue quartile. A Kruskal–Wallis test confirmed that the differences across revenue quartiles were statistically significant, with a test statistic of 105.933 and a p-value of 8.23×10^{-23} .

Revenue Quartile	Number of Companies	Median Weighted Degree
Q1 lowest revenue	92	7.0
Q2	91	13.0
Q3	91	36.0
Q4 highest revenue	91	60.0

Table 5: Weighted degree by revenue quartile.

A similar pattern appeared for market capitalization. Median weighted degree increased from 11.0 in the lowest market-cap quartile to 97.0 in the highest market-cap quartile. The Kruskal–Wallis test was statistically significant, with a statistic of 31.486 and a p-value of 6.71×10^{-7} .

Market-Cap Quartile	Number of Companies	Median Weighted Degree
Q1 lowest market cap	31	11.0
Q2	31	39.0
Q3	30	81.5
Q4 highest market cap	31	97.0

Table 6: Weighted degree by market capitalization quartile.

6.3 Interpretation

The statistical results support the hypothesis that larger companies are more visible in financial news and more central in the company co-mention network. Companies with higher revenue and market capitalization tend to appear in more articles and have higher weighted degree. This suggests that the structure of the news network is partly shaped by company size and market visibility.

The analysis therefore shows a clear relationship between company size and network position.

7 Financial Enrichment with yfinance

We also enriched the company network with financial market data using the Python package `yfinance`. This section introduces a different data source and prepares the project for later financial hypothesis testing.

The graph node dataset was merged with article counts and ticker information. The filtered graph contained 1,172 nodes. After extracting non-missing unique ticker symbols, 1,131 tickers were sent to yfinance. The notebook attempted to download historical price data these tickers over the period from 2017-01-01 to 2019-12-31. This window includes one year before and one year after the 2018 article dataset.

For each ticker with sufficient data, daily log returns were calculated as:

$$r_t = \log(P_t) - \log(P_{t-1}), \quad (7)$$

where P_t is the adjusted closing price at time t .

The enrichment process then computed several financial indicators, including the number of return observations, annualized volatility, cumulative return, mean daily return, etc... Annualized volatility was also calculated from daily returns.

Some tickers produced missing data or warnings, especially for companies that were delisted or did not have full Yahoo Finance coverage during the 2017–2019 period. Nevertheless, the enriched data creates a useful bridge between the financial news network and market behaviour.

8 Co-Mention Edges and Stock Return Co-Movement

This section tests whether the company co-mention graph is related to stock-market co-movement. The financial question is whether companies that are mentioned together in news articles also tend to have more correlated stock returns.

The intuition is that a co-mention edge may capture a real economic or market relationship between two companies. For example, two firms may be competitors, belong to the same sector, be exposed to the same macroeconomic shocks, or be jointly discussed by investors and financial journalists. If this is true, co-mentioned companies should have higher return correlation than randomly selected company pairs.

8.1 Dataset and Methodology

The analysis used three main inputs: the co-mention edge table, the company financial table, and the daily return panel for 2017–2019. Each co-mention edge was converted into a ticker pair. Pairs were kept only when both companies had valid stock tickers and enough overlapping daily returns.

For each valid pair of stocks, the Pearson correlation of daily log-returns was computed. This produced one return-correlation value per company pair. The edge group contained company pairs that were connected in the co-mention graph. The control group contained randomly sampled non-edge pairs from the same ticker universe.

The null hypothesis was that co-mentioned company pairs and random non-edge pairs have the same average return correlation. The alternative hypothesis was that co-mentioned company pairs have higher average return correlation.

Because correlation values are bounded between -1 and 1, the correlations were transformed using Fisher’s transformation before the statistical test:

$$z = \operatorname{arctanh}(r), \quad (8)$$

where r is the Pearson correlation. This transformation makes correlations more suitable for mean-based comparison.

A Welch two-sample t-test was used as the main test because the normality and equal-variance assumptions were not fully satisfied. A Mann–Whitney test was also reported as a robustness check.

8.2 Results

The final test used 4,159 co-mention edge pairs and 4,159 random non-edge pairs. Co-mentioned pairs had a higher average daily return correlation than random non-edge pairs.

Group	N	Mean Correlation	Median Correlation	Std. Dev.
Co-mention edge pairs	4,159	0.302	0.281	0.169
Random non-edge pairs	4,159	0.195	0.191	0.128

Table 7: Return correlation comparison between co-mention edges and random non-edges.

The Welch two-sample t-test strongly rejected the null hypothesis, with a test statistic of 31.908 and a p-value below 0.001. The Mann–Whitney robustness check also produced a p-value below 0.001. This means that the difference is statistically significant and unlikely to be explained by random sampling variation alone.

The second part of the analysis tested whether stronger co-mention relationships were associated with stronger stock co-movement. Edge strength was measured by the co-mention weight, meaning the number of articles in which two companies appeared together.

The Pearson correlation between log co-mention weight and Fisher-transformed return correlation was 0.293, with a p-value of 2.4×10^{-83} . Spearman correlation was added as a robustness check because it is based on ranks and is less sensitive to the exact linear form of the relationship and showed a positive association, with $\rho = 0.215$ and a p-value of 9.01×10^{-45} .

8.3 Sector Robustness Check

A possible concern is that the higher return correlation of co-mentioned firms may simply come from sector effects. Companies in the same sector naturally tend to move together because they are exposed to similar economic conditions, regulation, supply chains, and investor expectations.

To check this, the co-mention edges were separated into same-sector and cross-sector edges. Same-sector edges connect two companies assigned to the same broad sector, while cross-sector edges connect companies from different sectors.

Same-sector co-mention edges had the highest average return correlation, which is expected. However, cross-sector edges also had a mean correlation of 0.263, which is still higher than the

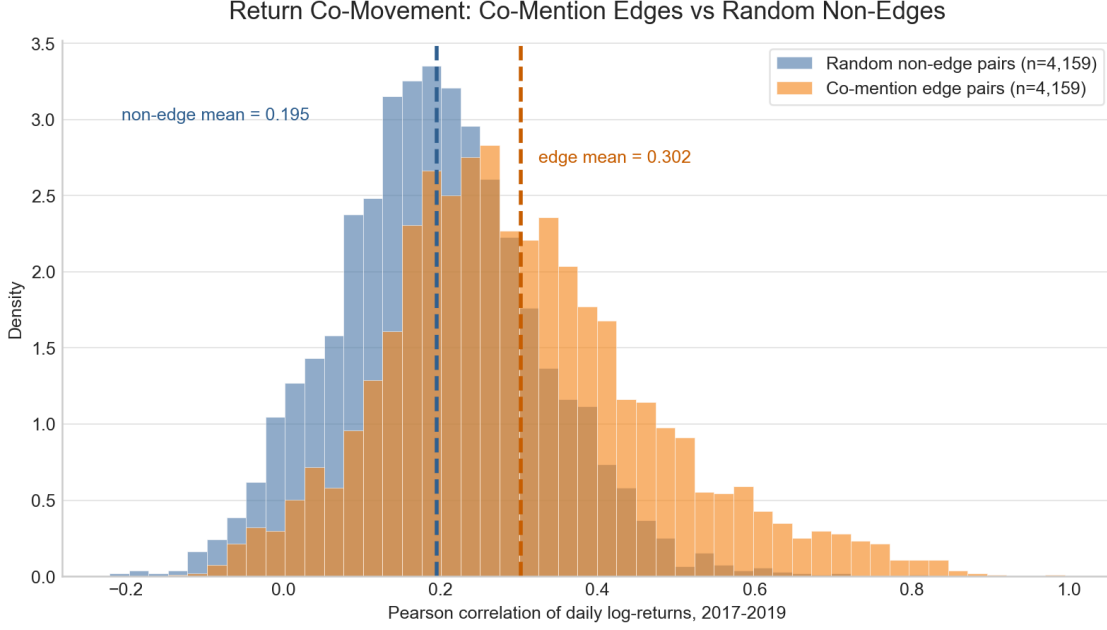


Figure 5: Distribution of return correlations for co-mention edge pairs and random non-edge pairs.

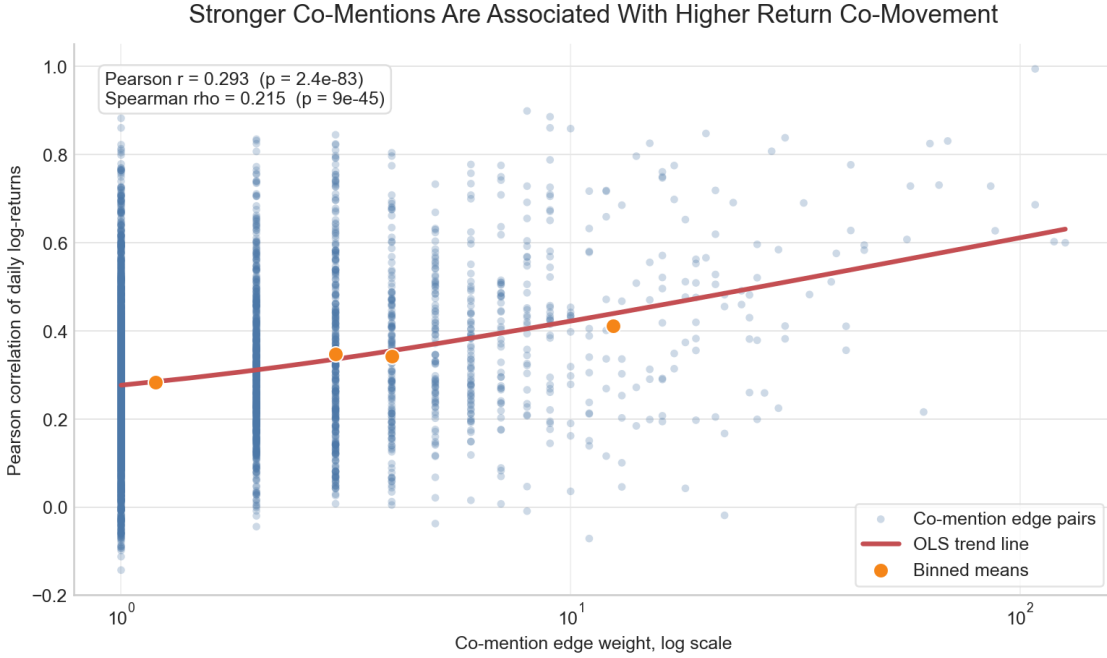


Figure 6: Relationship between co-mention edge weight and stock return co-movement.

random non-edge mean of 0.195. This suggests that the co-mention effect is not only driven by firms belonging to the same sector. The news graph also captures broader market, supply-chain, financial, or attention-based links between companies from different sectors.

Edge Type	N	Mean Correlation	Median Correlation	Std. Dev.
Cross-sector edges	2,830	0.263	0.255	0.140
Same-sector edges	1,329	0.387	0.371	0.194

Table 8: Return correlation for same-sector and cross-sector co-mention edges.

8.4 Interpretation

The results indicate that co-mentioned companies tend to have higher stock-return co-movement than randomly selected company pairs. This suggests that the co-mention graph is not only capturing textual proximity, but also some form of economic or market-related proximity between firms.

The edge-weight analysis gives a similar conclusion. Pairs that appear together in more articles generally have higher return correlations. The relationship is positive but not perfect, which is expected because stock returns are influenced by many factors that are not captured by news co-mentions alone.

The sector robustness check also helps qualify the result. Same-sector edges have the highest correlations, but cross-sector edges remain more correlated than random non-edge pairs. Therefore, sector membership explains part of the effect, but it does not fully account for it. Co-mentions also seem to capture broader relationships between companies, such as shared exposure to market events, supply-chain links, or common investor attention.

9 Graph Centrality and Stock Volatility

This section studies whether a company’s position in the co-mention graph is related to its stock volatility. The financial question is whether companies that are more central in the news network have different levels of stock-market risk.

The main graph variable is PageRank centrality. In this project, a company has high PageRank when it is connected to other important companies in the co-mention graph. The financial variable is annualised volatility, computed from daily log-returns over the 2017–2019 period.

9.1 Dataset and Methodology

The analysis used the precomputed graph centrality table and the financial enrichment table. The centrality table contains Louvain clusters, industry groups, weighted degree, betweenness, and PageRank. The financial table contains stock-return indicators such as annualised volatility, cumulative return, and a Sharpe proxy.

After merging the two tables, the final dataset contained 727 companies with both graph centrality and valid stock volatility data.

Companies were divided into three equal-sized groups according to their PageRank value: low centrality, middle centrality, and high centrality. This allowed us to compare average volatility across the three groups.

The null hypothesis was that the three PageRank groups have the same mean annualised volatility:

$$H_0 : \mu_{\text{low}} = \mu_{\text{middle}} = \mu_{\text{high}}. \quad (9)$$

The alternative hypothesis was that at least one group has a different mean volatility.

The normality assumption was checked using the Shapiro–Wilk test, and equality of variances was checked using Levene’s test. Since the volatility distributions were not normally distributed and the variances were unequal, Welch ANOVA was used as the main test. Kruskal–Wallis was also reported as a non-parametric robustness check.

9.2 Results

The descriptive statistics show that annualised volatility decreases from the low PageRank group to the high PageRank group.

PageRank Group	N	Mean Volatility	Median Volatility	Std. Dev.
Low centrality	242	0.377	0.311	0.228
Middle centrality	242	0.362	0.301	0.236
High centrality	243	0.283	0.245	0.156

Table 9: Annualised volatility by PageRank centrality group.

The assumption checks rejected normality in all three groups. Levene’s test also rejected equal variances. Therefore, Welch ANOVA was used as the main omnibus test.

Test	Statistic	p-value
Classical ANOVA	13.946	1.14×10^{-6}
Welch ANOVA	17.871	3.34×10^{-8}
Kruskal–Wallis robustness check	53.740	2.14×10^{-12}

Table 10: Comparison of annualised volatility across PageRank groups.

The Welch ANOVA result is statistically significant, meaning that average volatility differs across the three centrality groups.

A direct comparison between the high-centrality and low-centrality groups was then performed. The high-centrality group had a mean annualised volatility of 0.283, while the low-centrality group had a mean annualised volatility of 0.377. The Welch two-sample t-test gave a statistic of -5.267 and a p-value of 2.21×10^{-7} . The Mann–Whitney robustness check also gave a significant result, with a p-value of 1.61×10^{-11} .

Finally, PageRank was also treated as a continuous variable. The Pearson correlation between log PageRank and volatility was -0.206, with a p-value of 2.05×10^{-8} . The Spearman rank correlation was -0.283, with a p-value of 6.81×10^{-15} .

Comparison	Statistic	p-value
Welch two-sample t-test	-5.267	2.21×10^{-7}
Mann-Whitney robustness check	19005.0	1.61×10^{-11}

Table 11: High-centrality versus low-centrality volatility comparison.

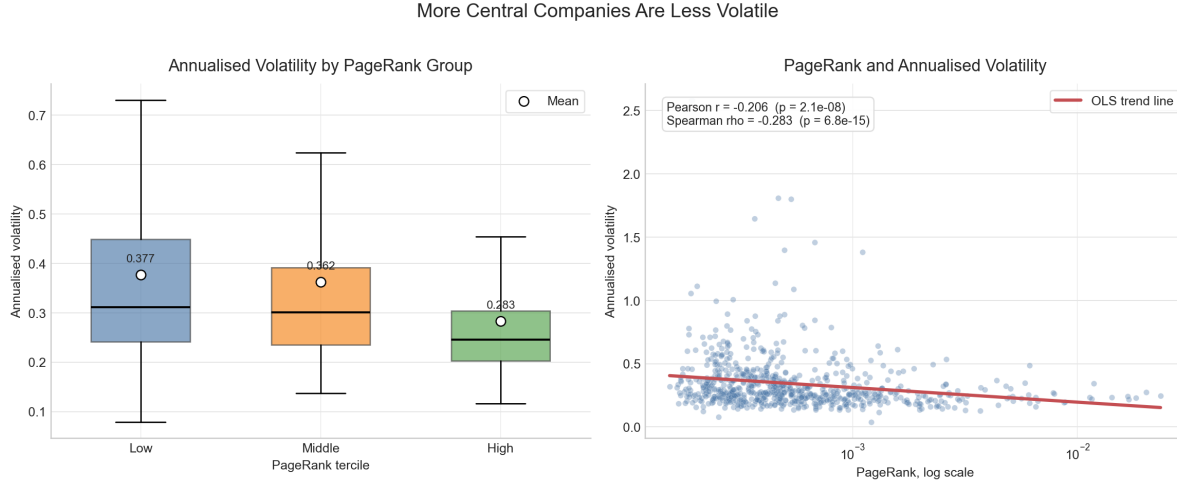


Figure 7: Annualised volatility by PageRank group and continuous PageRank-volatility relationship.

9.3 Interpretation

The results indicate a negative relationship between PageRank centrality and stock volatility. Companies that are more central in the co-mention graph tend to have lower annualised volatility.

This result is not necessarily surprising once the meaning of centrality is considered. In this graph, highly central companies are often large and highly visible firms, such as major technology companies, banks, and multinational corporations. These companies are mentioned frequently and are connected to many other important firms, but they may also be more stable and diversified than smaller or less central companies.

The grouped tests and the continuous correlation checks point in the same direction. The effect is statistically significant, but the correlation is moderate rather than strong. This means that PageRank is related to volatility, but it is only one of several factors that explain stock risk.

10 Mention Spikes and Next-Week Stock Reactions

This section studies whether unusually high media attention is followed by abnormal stock behaviour. The financial question is whether a company experiences larger stock movements after a week in which it is mentioned unusually often in financial news.

The idea is different from the previous graph-level tests. Instead of comparing company pairs or graph centrality groups, this analysis focuses on each ticker's own time series of news mentions. A mention spike is defined relative to the company's usual level of media attention.

10.1 Dataset and Methodology

The analysis used the article-company mention table, the cleaned company reference dataset, and the daily return panel. Article-level company names were matched to stock tickers using the same alias-normalization logic used earlier in the project. After matching, the dataset contained 17,282 article-ticker rows with valid stock-return data.

Mentions were then aggregated by ticker and week. Daily log-returns were also aggregated into weekly log-returns by summing daily returns within each week. This produced a weekly panel of mention counts and stock returns.

For each ticker, spike weeks were defined as weeks where the number of mentions was in the top decile of that ticker’s positive weekly mention counts. Several filters were applied to keep only tickers with enough information for comparison. A ticker needed at least 20 total mentions, at least 3 spike weeks, and at least 20 normal weeks.

For each ticker, two quantities were computed:

$$\text{Absolute return effect} = \overline{|r_{t+1}|}_{\text{spike}} - \overline{|r_{t+1}|}_{\text{normal}}, \quad (10)$$

and

$$\text{Signed return effect} = \overline{r_{t+1}}_{\text{spike}} - \overline{r_{t+1}}_{\text{normal}}. \quad (11)$$

The first quantity measures whether prices move more after mention spikes, regardless of direction. The second quantity tests whether mention spikes are followed by systematically positive or negative returns.

The statistical test was performed across tickers, using one effect value per ticker. This avoids treating many weeks from the same company as independent observations. Since the final sample contained 84 tickers and the effect distributions were approximately normal according to the Shapiro–Wilk test, one-sample t-tests were used. Wilcoxon signed-rank tests were also reported as robustness checks.

10.2 Results

After applying the filters, 84 tickers were kept for the event-study analysis.

The first test examined whether mention spikes were followed by larger absolute returns in the next week. The mean ticker-level absolute-return effect was 0.0057, and the median effect was 0.0067. The one-sample t-test rejected the null hypothesis of zero effect, with a p-value of 0.0010. The Wilcoxon robustness check gave a similar conclusion, with a p-value of 0.0007.

The second test examined signed next-week returns. The mean ticker-level signed-return effect was -0.0052, and the median effect was -0.0048. The two-sided one-sample t-test gave a p-value of 0.0299, and the Wilcoxon robustness check gave a p-value of 0.0221.

Measure	Value
Number of tickers	84
Mean absolute-return effect	0.0057
Median absolute-return effect	0.0067
One-sample t-test statistic	3.185
One-sample t-test p-value	0.0010
Wilcoxon robustness p-value	0.0007

Table 12: Test of next-week absolute-return effects after mention spikes.

Measure	Value
Number of tickers	84
Mean signed-return effect	-0.0052
Median signed-return effect	-0.0048
One-sample t-test statistic	-2.209
One-sample t-test p-value	0.0299
Wilcoxon robustness p-value	0.0221

Table 13: Test of next-week signed-return effects after mention spikes.

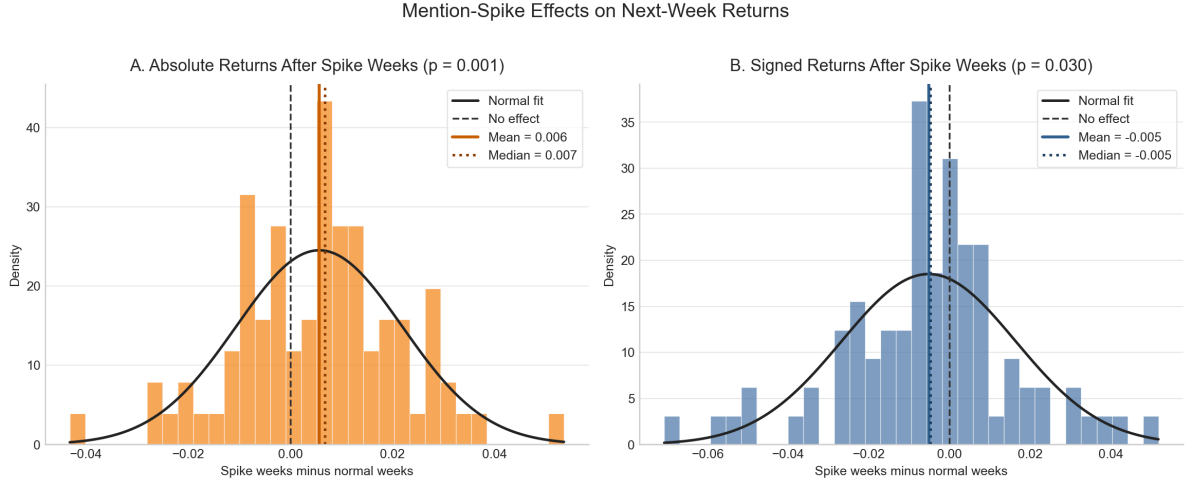


Figure 8: Ticker-level effects of mention spikes on next-week absolute and signed returns.

10.3 Interpretation

The results suggest that unusually high media attention is followed by larger next-week stock movements. The absolute-return test is the main result because it measures the size of the price reaction without assuming a direction. This test is statistically significant and robust to the non-parametric check. This means that, across the 84 tickers that pass the filters, stocks tend to move more in the week after a mention spike than after a normal week.

The signed-return result is also statistically significant, but it should be interpreted more carefully. The negative mean suggests that mention spikes were followed, on average, by slightly lower

next-week returns. However, this does not necessarily mean that media attention causes prices to fall. Mention spikes can happen during negative events, earnings surprises, litigation, market stress, or other situations where the news itself is already reflecting bad information.

Overall, the evidence is stronger for increased absolute returns after mention spikes than for a clear directional return effect, which suggests that bursts of media attention capture periods of higher market uncertainty or information arrival.

11 News Mentions, Company Visibility, and Stock Reaction

This section tests whether less-mentioned companies react more strongly to news mentions than highly mentioned companies. The financial question is whether a news mention has a larger effect on companies that receive less regular media attention.

The intuition is that highly visible companies are already followed by many investors, analysts, and journalists. Therefore, one additional news mention may not contain much unexpected information. For less visible companies, however, a news mention may represent a stronger attention shock and may be followed by a larger stock-price reaction.

11.1 Dataset and Methodology

The analysis used the article-company mention table, the cleaned company reference dataset, the graph node table, article counts, and daily stock returns. Company names from articles were matched to tickers using the same alias-normalization procedure used earlier in the project.

The first step was to build ticker-date mention events. Each event corresponds to one ticker being mentioned on one article date. Duplicate ticker-date pairs were removed, so if the same company appeared in several articles on the same date, it was counted as one mention event for that date. This produced 9,554 unique ticker-date mention events.

Companies were then classified by article count. Article count was used as a proxy for company visibility. Companies in the bottom third of article count were classified as low-article-count companies, while companies in the top third were classified as high-article-count companies. The middle third was excluded from the main comparison in order to focus on the clearest contrast.

For each ticker-date event, returns were aligned with the trading calendar. The event window used the previous trading day and the next trading day:

$$r(-1) = \text{return on the previous trading day}, \quad (12)$$

$$r(+1) = \text{return on the next trading day}. \quad (13)$$

The main reaction variable was defined as:

$$\Delta|r| = |r(+1)| - |r(-1)|. \quad (14)$$

A positive value means that the absolute stock movement was larger after the news mention than before it. This variable measures the size of the price reaction, not its direction.

The analysis was then aggregated to the ticker level. For each ticker, the average value of $\Delta|r|$ was computed across all its mention dates. This step was important because otherwise very frequently mentioned companies would contribute many more observations than less frequently mentioned companies. Only tickers with at least three mention events were kept.

The final statistical test compared the ticker-level average reaction of low-article-count companies with that of high-article-count companies.

11.2 Results

The event-window dataset contained 8,105 usable events for low- and high-article-count companies. At the event level, low-article-count companies had a larger average increase in absolute returns after news mentions.

Group	Events	Mean $\Delta r $	Median $\Delta r $
Low article count	710	0.0087	0.0014
High article count	7,395	0.0002	0.0000

Table 14: Event-level absolute-return changes after news mentions.

After aggregating to one row per ticker and requiring at least three events per ticker, the final sample contained 152 low-article-count companies and 284 high-article-count companies.

The main test compared the ticker-level mean absolute-return change between the two groups. Since the normality and equal-variance assumptions were not satisfied, Welch’s two-sample t-test was used as the main test. Mann–Whitney was also reported as a robustness check.

Measure	Low Article Count	High Article Count
Number of tickers	152	284
Mean $\Delta r $	0.0061	0.0003
Median $\Delta r $	0.0036	0.0001

Table 15: Ticker-level post-news absolute-return reaction by article-count group.

The Welch two-sample t-test gave a statistic of 3.438 and a p-value of 0.000372. The Mann–Whitney robustness check also supported the result, with a p-value of 0.000006.

Test	Statistic	p-value
Welch two-sample t-test	3.438	0.000372
Mann–Whitney robustness check	27073.0	0.000006

Table 16: Comparison of ticker-level reactions between low- and high-article-count companies.

Within-group tests were also performed to check whether the average reaction was significantly

greater than zero within each group. For high-article-count companies, the effect was not statistically significant. For low-article-count companies, the effect was statistically significant.

Group	Mean $\Delta r $	t-test p-value	Wilcoxon p-value
Low article count	0.0061	0.000177	0.000005
High article count	0.0003	0.224052	0.262395

Table 17: Within-group tests of post-news absolute-return reactions.

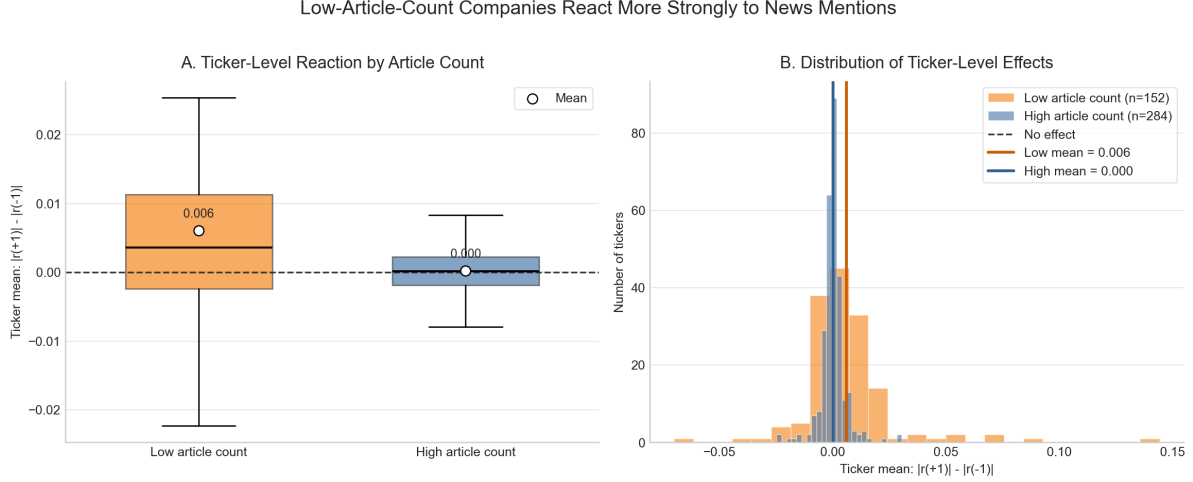


Figure 9: Ticker-level post-news reaction by article-count group.

11.3 Methodological Limitation

This analysis should be interpreted as a simplified short-window reaction test rather than a full abnormal-return event study. The reaction variable compares the absolute stock return after a news mention with the absolute stock return before the mention:

$$\Delta|r| = |r(+1)| - |r(-1)|. \quad (15)$$

This measure is useful because it gives a simple indication of whether the stock moved more after the mention date than before it. However, it has two important limitations.

First, the measure is not adjusted for market-wide movements. If the entire market was more volatile on the day after the article, then a higher value of $|r(+1)|$ may reflect general market conditions rather than a company-specific reaction to the news mention.

Second, the baseline uses only one pre-event trading day. This is noisy because a single day may not represent the normal volatility of the company.

Therefore, the results should not be interpreted as causal evidence that the news mention itself moved the stock price. They should instead be read as evidence of an association between news mentions and larger short-window stock movements, especially for low-article-count companies.

11.4 Interpretation

The results indicate that low-article-count companies are associated with stronger short-window stock movements around news mentions than high-article-count companies. The average post-news absolute-return increase is much larger for low-article-count firms, and the difference between the two groups is statistically significant.

The within-group tests also support this interpretation. For high-article-count companies, the average reaction is close to zero and not statistically significant. For low-article-count companies, the reaction is positive and significant.

This suggests that a news mention is more informative for companies that are rarely mentioned. For highly covered companies, an additional article is less surprising and does not meaningfully change next-day price movement. For low-article-count companies, a mention is a stronger attention shock and is followed by a larger stock reaction.

This result should still be interpreted as an association rather than a causal effect. The news mention may coincide with an underlying event, such as earnings, litigation, or sector news, which itself causes the stock reaction. However, the evidence is consistent with the idea that media attention has a stronger market association for less visible companies.

12 Conclusion

In this project, we studied whether company mentions in financial news can give useful information about relationships between firms and about market behaviour. Starting from financial news articles, we extracted company names, matched them to listed companies, and built a co-mention graph where two companies are connected when they appear in the same article.

The graph analysis showed that the network has a meaningful structure. The most central companies were mainly large and well-known firms, especially in technology and finance. Louvain communities were also partly aligned with industry groups, and this alignment became stronger when weak co-mention edges were removed. This suggests that repeated co-mentions are more informative than isolated co-mentions.

The financial tests gave similar evidence. Companies connected in the co-mention graph had higher return correlations than random company pairs. Also, pairs with stronger co-mention weights tended to have higher return co-movement. This means that the graph is not only describing the text, but also captures some relationships that are visible in stock returns.

The centrality-volatility analysis showed that more central companies were less volatile on average. This was an interesting result because one could expect central companies to be more exposed to news. In our data, centrality seems instead to capture large and established companies, which are more visible but also more stable.

The mention-spike analysis showed that weeks with unusually high attention were followed by larger next-week absolute returns. This does not mean that news attention directly causes the movement, but it suggests that mention spikes are associated with periods where more information

is arriving to the market.

Finally, the short-window reaction test showed that low-article-count companies had larger post-news absolute-return changes than high-article-count companies. This result is consistent with the idea that a news mention may be more important for companies that are usually less visible. However, this part should be interpreted carefully because the test does not use a full abnormal-return event-study framework.

Overall, the project shows that financial news co-mentions can be useful for studying company relationships and market reactions. The statistical results should therefore be interpreted as evidence of associations rather than direct causal effects. A natural extension of this work would be to refine the event-study design by using market-adjusted abnormal returns and longer pre-event estimation windows.

Github Link:

<https://github.com/NicolasBrj/Modal-Nicolas-El-Bourji-Jean-Paul-Aldawaliby.git>